

Ushtrime ND10dh

2.1.22 Sa është temperatura e përzierjes së krijuar nga $m_1=0,5\text{kg}$ ujë në temperaturë

$t_1=20^\circ\text{C}$ dhe $m_2=0,3\text{kg}$ alkool me temperaturë $t_2=80^\circ\text{C}$?

Zgjidhje:

Të dhënat:

$$m_1=0,5\text{kg} \text{ (H}_2\text{O)}$$

$$c_1=4190 \frac{\text{J}}{\text{kg}^\circ\text{C}}$$

$$t_1=20^\circ\text{C}$$

$$m_2=0,3\text{kg} \text{ (alkool)}$$

$$c_2=2400 \frac{\text{J}}{\text{kg}^\circ\text{C}}$$

$$t_2=80^\circ\text{C}$$

$$t_e=?$$

$$Q_1=Q_2$$

$$Q_1=m_1 c_1 \Delta t$$

$$\Delta t \neq \Delta t'$$

$$Q_2=m_2 c_2 \Delta t'$$

$$\Delta t = t_e - t_1, \quad t_e > t_1$$

$$\Delta t' = t_2 - t_e, \quad t_2 > t_e$$

$$m_1 c_1 \Delta t = m_2 c_2 \Delta t'$$

$$m_1 c_1 (t_e - t_1) = m_2 c_2 (t_2 - t_e)$$

$$m_1 c_1 t_e - m_1 c_1 t_1 = m_2 c_2 t_2 - m_2 c_2 t_e$$

$$m_1 c_1 t_e + m_2 c_2 t_e = m_1 c_1 t_1 + m_2 c_2 t_2$$

$$t_e (m_1 c_1 + m_2 c_2) = m_1 c_1 t_1 + m_2 c_2 t_2 \quad / : (m_1 c_1 + m_2 c_2)$$

$$t_e = \frac{m_1 c_1 t_1 + m_2 c_2 t_2}{m_1 c_1 + m_2 c_2} = \frac{0,5\text{kg} \cdot 4190 \frac{\text{J}}{\text{kg}^\circ\text{C}} \cdot 20^\circ\text{C} + 0,3\text{kg} \cdot 2400 \frac{\text{J}}{\text{kg}^\circ\text{C}} \cdot 80^\circ\text{C}}{0,5\text{kg} \cdot 4190 \frac{\text{J}}{\text{kg}^\circ\text{C}} + 0,3\text{kg} \cdot 2400 \frac{\text{J}}{\text{kg}^\circ\text{C}}}$$

$$t_e = \frac{41900 \text{ J} + 57600 \text{ J}}{2095 \frac{\text{J}}{^\circ\text{C}} + 720 \frac{\text{J}}{^\circ\text{C}}} = \frac{99500 \cancel{\text{J}}}{2815 \cancel{\frac{\text{J}}{^\circ\text{C}}}} = 35,346^\circ\text{C}$$

$$\boxed{t_e \approx 35,35^\circ\text{C}}$$